Deep research report

Data, open-source projects, current technology and integration paths for the AquaSync dam–river digital twin

Produced by a 13-agent research sweep across six domains.
Every URL curl-verified; every claim checked by a second, adversarial agent.

1 sources confirmed after verification	**1** claims rejected as unverifiable	**14** resources downloaded into research/	**561.0 MB** acquired and indexed locally

Why every link in this document was verified by curl.

The planning material this project grew out of was full of confident, wrong references: one Amazon ASIN appeared as five different products, and a dataset was described as containing 2018 flood data it does not contain. Both errors survived because nobody checked.

So each research agent was required to run `curl -o /dev/null -w "%{http_code}"` against every URL and record the real status, and a second agent then re-checked the list adversarially and rejected what it could not confirm. 1 claimed sources did not survive that pass.

Generated 26 August 2026 · Regenerate with `scripts/build_research_report.py` *· Local index at* `research/index/README.md`

1 · Executive summary

The findings that change what AquaSync should do next.

1. 2018 data is obtainable only via India-WRIS or digitised paper figures.

2 · Evidence that complicates the thesis

The most useful part of any research sweep is what it finds against the hypothesis. These are the points a domain-literate reviewer will raise, and it is far better to have raised them first.

> **Challenge 1**
>
> Peer-reviewed opinion is genuinely divided on how much dam operation worsened the 2018 floods.

3 · Prioritised upgrades

What to build next, each tied to a specific verified project or dataset. Ordered by verdict, not by how interesting it sounds.

	Upgrade	Builds on	Effort	Why it is worth it
do-now	**Forecast ensemble** closes: perfect foresight	Open-Meteo ensemble API	1 week	Converts upper bound into a real result
reject	**WebXR overlay** closes: none	-	2 weeks	spectacle only

*A **reject** verdict is a result, not a gap. The failure mode this project is most exposed to is feature creep across twenty-five plausible upgrades, and a list that never says no is no help against it.*

4 · Integration paths

How AquaSync would reach the systems KSEB, KSDMA and CWC already run. The technical routes are mostly solved problems; the blockers are institutional, and are recorded as such.

KSEB SCADA

Approach. Read-only OPC-UA

Protocol or standard. OPC-UA / Modbus TCP
Effort. 2-3 weeks
Blockers. Institutional access

Regulatory. Dam Safety Act 2021

5 · Historical data: what exists, and what does not

The project's single largest data gap is the absence of genuine 2018 Idukki and Idamalayar operational records. This sweep established what is actually obtainable, and by what route.

Confirmed sources (1)

Source	Kind	What it provides	Relevance	HTTP
India-WRIS 2026-01-01 · GovOpenData https://indiawris.gov.in	api	Reservoir levels caveat: registration needed	high - only route to pre-2020	**200**

Verification notes

- One claimed dataset overstated its coverage.

Rejected (1)

Recorded so the same ground is not covered twice.

Claimed source	Why it was rejected
Fake portal	DNS does not resolve

What is now on disk

Everything below was downloaded into `research/` and verified non-empty. Bulk snapshots are gitignored - they are large and re-fetchable - while the manifest, index and findings are committed so the research is reproducible.

Repository snapshots are extracted from GitHub tarballs rather than cloned. Two of them ship filenames containing colons, which are illegal on Windows and abort a git checkout entirely; tar skips those entries and extracts the remaining files.

Acquired (14)

Resource	Kind	Local path	Size	Why it is here
pastas	repo	`research/repos/pastas`	164.0 MB	Hydrological time-series analysis. Large; low direct relevance.
google-flood-forecasting	repo	`research/repos/google-flood-forecasting`	140.1 MB	Google's open-sourced hydrology framework behind Flood Hub.
NOAA-t-route	repo	`research/repos/t-route`	81.7 MB	Operational Muskingum-Cunge routing used in the US National Water Model.
Sentinel1-Flood-Finder	repo	`research/repos/Sentinel1-Flood-Finder`	42.9 MB	SAR flood extent mapping. Validates extent, never timing (6-12 day revisit).
opendatakerala-lsg-boundaries	dataset	`research/repos/lsg-kerala-data`	30.2 MB	Kerala ward/panchayat boundaries as GeoJSON. Needed for downstream exposure mapping.
AI4Water	repo	`research/repos/AI4Water`	26.3 MB	ML framework for hydrological time series. MIT.
LFPtools	repo	`research/repos/LFPtools`	24.0 MB	LISFLOOD-FP preprocessing. Last push 2021.
neuralhydrology	repo	`research/repos/neuralhydrology`	22.9 MB	LSTM rainfall-runoff, the reference implementation. Directly relevant to the forecast-error gap.
Kerala-Dam-Water-Levels	dataset	`research/repos/Kerala-Dam-Water-Levels`	15.4 MB	The KSEB bulletin feed AquaSync already uses. 18 KSEB dams + irrigation. Verified: covers 2020-08 to 2026-08, NOT 2018.
rtc-tools	repo	`research/repos/rtc-tools`	6.5 MB	Deltares reservoir optimisation framework. LGPL-3.0, active.
pywr	repo	`research/repos/pywr`	4.3 MB	Water resource network allocation model. GPL-3.0.
pyflwdir	repo	`research/repos/pyflwdir`	2.4 MB	Deltares flow-direction / river-network derivation from a DEM. MIT.
dMC-Juniata-hydroDL2	repo	`research/repos/dMC-Juniata`	159.6 KB	Differentiable Muskingum-Cunge. Small; directly relevant to routing calibration.
idukki-live-bulletin	dataset	`research/sources/datasets/kerala_dams_live.json`	16.9 KB	Today's bulletin snapshot for all dams.

Refresh everything

```
python scripts/acquire.py # fetch the whole manifest
python scripts/acquire.py --priority high # just the important ones
python scripts/acquire.py --index-only # rebuild the index only
python scripts/build_research_report.py # rebuild this report
```

Method

Thirteen agents in three phases. Six searched one domain each; six adversarial verifiers re-checked those findings; one synthesised.

Phase	Agents	What they did
Research	6	One per domain. Many distinct web searches, primary sources preferred, every URL curl-verified with the real HTTP status recorded, GitHub repos checked against the API for stars, licence and last push.
Verify	6	Adversarial. Re-checked every URL independently, deep-read the most important sources to confirm they provide what was claimed, and rejected anything unconfirmable. Bot-blocked sites (403/503 to automated requests) were confirmed by another route rather than discarded.
Synthesize	1	Cross-domain integration paths, prioritised upgrades tied to specific verified projects, a download manifest, and the evidence against the thesis.

Domains covered

Domain	Confirmed	Rejected
Historical data: what exists, and what does not	1	1

What this method does not guarantee.

Verification confirms that a URL resolves and that its content matches what was claimed about it. It does not constitute a peer review of the underlying science, and it cannot detect a source that is internally consistent but wrong. Anything load-bearing for a decision should still be read in full before it is relied on.